

摘要

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Keywords: keyword1; keyword2

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1 Introduction

1.1 Problem Background

In recent years, generative artificial intelligence has become deeply integrated into learning and work, supporting writing, programming, design, information retrieval, and content creation while reshaping workflows in many roles. Employer surveys indicate that demand for key skills will shift significantly in the coming years. Skills gaps have become a major challenge for companies, and many now regard employee reskilling and upskilling as a central strategy.

Different professions are affected in different ways. In some occupations, a large share of tasks can be supported or partially undertaken by generative AI, while others rely more on on-site operations or interpersonal collaboration. Task-level estimates suggest that the jobs of roughly one quarter of workers worldwide will be affected to some degree, but the more common outcome is the reorganization and upgrading of work rather than the direct replacement of positions. This creates urgent pressure on education systems to adjust curricula and assessment methods so that students learn to use generative AI to enhance their capabilities while limiting negative impacts, at a time when many countries still lack adequate regulation and data privacy protection and educational institutions have not yet formed reliable practices for integrating these tools into teaching and research.

At the same time, the spread of generative AI brings resource and environmental pressures, as model training and deployment consume large amounts of electricity and hardware cooling increases water use, affecting local water supplies and ecosystems. Against this backdrop, this study focuses on linking changes in the world of work with educational decision-making through a data-driven approach. We analyze task changes and evolving skill demands across different occupations, then translate these findings into actionable adjustment plans for schools and programs, covering curriculum content, practical training, assessment design, faculty allocation, and risk management such as usage guidelines and rules for attribution and acknowledgment. Through multi-metric evaluation and uncertainty analysis, we aim to provide more robust recommendations and to build reusable decision-making frameworks for a wide range of training programs.

1.2 Restatement of the Problem

Problem 1: Design a clear, data-driven method that, against the backdrop of Gen-AI-induced changes in job tasks and skill demands, first screens and organizes a large set of occupations, and then selects a small number of representative occupations for more in-depth, targeted studies at the institutional level. We begin by using an occupation-year panel to build a composite index covering six dimensions: labor demand, income returns, skill structure, technology use, adaptability, and institutional environment. We then clean and standardize the data and apply an entropy-weight method to obtain annual composite scores for each occupation. Next, we use the K-means algorithm to cluster occupations based on recent score trends and identify several core occupation types. Finally, we forecast the development trends of each type and select one target occupation each from STEM, trades,

Placeholder for a chart summarizing projected skill-demand shifts and the share of firms prioritizing reskilling/upskilling.

图 1: Projected shifts in key skill demand and employer emphasis on reskilling/upskilling (survey-based).

Placeholder for a chart showing task-level exposure to generative AI across occupations (e.g., histogram/stacked bars by occupation group).

图 2: Estimated distribution of task-level exposure to generative AI across occupations, highlighting heterogeneous impacts.

and arts for subsequent analysis.

1.3 Literature Review

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1.4 Our Work

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2 Assumptions and Justifications

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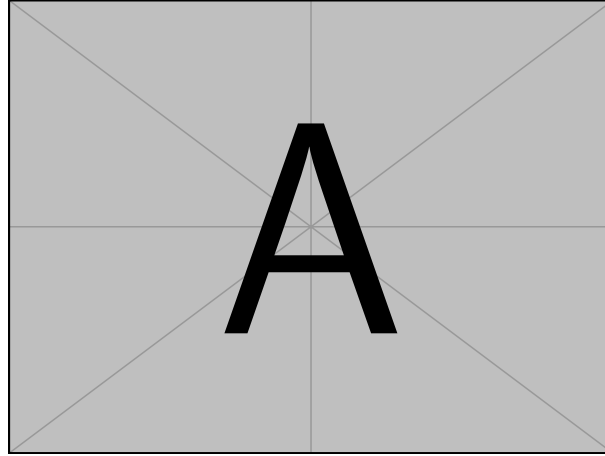


图 3: The name of figure

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$$\arcsin \theta = \iiint_{\varphi} \lim_{x \rightarrow \infty} \frac{n!}{r!(n-r)!} \quad (1)$$

3 Data Sources and Preprocessing

4 Notations

For each indicator X , the subscript i denotes the i -th career. The superscript (t) denotes the year index. i.e. $X_i^{(2018)}$ is the value of that indicator for career i in 2018.

5 Multi-Objective Optimization Model

5.1 Sub-Model I: Entropy-Weight-Based Career Screening and Representative Occupation Identification via K-Means

5.1.1 Indicator Selection and Data Preparation

Sub-Model I first develops a unified indicator system to describe how different careers have evolved over the past n years in the context of the ongoing development of generative AI (Gen-AI). Let $i = 1, \dots, N$ index careers, $j = 1, \dots, m$ index indicators, and $t \in t_1, \dots, t_n$ index years. For each career-year observation, the study compiles a vector of raw indicator values $x_{ij}^{(t)}$ spanning six dimensions: labor market demand, economic return, skill structure, dependence on technology, innovation and adaptability, and the institutional environment.

Data Processing. 为确保不同数据源和不同年份之间的可比性, 本文对原始职业样本开展两步清洗和缺失值处理, 先剔除就业人数低于 50,000 的职业, 避免小样本职业带来的数值大幅波动和评估结果不稳定, 再对缺失数据采用职业级阈值删除加同类别均值插补的处理策略, 若某职业在任意一项关键指标上的缺失比例超过 10%, 即判定其信息不足以支撑可靠评估并予以剔除, 若未超过则用该职业所属类别在同一年份的指标均值填补缺失值, 这套处理流程在控制数据噪声和维持样本规模之间实现了平衡, 能够为后续的聚类 and 综合评分提供一致且可复现的数据基础。

5.1.2 Entropy Weighting and K-Means Clustering

After normalization, the analysis proceeds with two core objectives. First, multi-dimensional indicators are aggregated into a single annual score so that observations remain comparable across both occupations and years. Second, a small set of occupational archetypes is identified that is statistically coherent and conceptually interpretable, thereby supporting subsequent decision-making.

To avoid subjective weighting, this study adopts the entropy weight method (EWM). EWM assigns larger weights to indicators that exhibit greater dispersion in the occupation-year panel and therefore contain more discriminative information. Conversely, indicators that remain relatively stable across occupations provide limited screening power and receive lower weights through the entropy mechanism. Let w_j denote the entropy-based weight of indicator j , computed from the full panel. The annual

表 1: Indicator notations for the 4 dimensions.

Symbol	Indicator	Meaning	Unit
Dim1: Labor Market Demand (LMD)			
$LMD1_i^{(t)}$	Employment size	Number of employed persons	
$LMD2_i^{(t)}$	Employment growth rate	Year-over-year employment growth rate	%
$LMD3_i^{(t)}$	Unemployment rate	Unemployment rate associated	%
$LMD4_i^{(t)}$	New job creation rate	Rate of newly created jobs	%
Dim2: Compensation and Economic Return (CER)			
$CER1_i^{(t)}$	Median annual wage	Income level	USD/year
$CER2_i^{(t)}$	Wage growth rate	Return growth	%
$CER3_i^{(t)}$	Income dispersion	Within-occupation wage inequality	index
$CER4_i^{(t)}$	Education return ratio	Income per training/education cost	ratio
$CER5_i^{(t)}$	Medium-to-high premium rate	The premium range from middle-income to high-income segments	index
Dim3: Skill Structure (SKS)			
$SKS1_i^{(t)}$	Share of high-skill positions	Cognitive/professional skill intensity	%
$SKS2_i^{(t)}$	Skill update frequency	Speed of skill iteration	1/year
Dim4: Technology Dependence (TED)			
$TED1_i^{(t)}$	Digital tool usage rate	Degree of technology penetration in daily tasks	%
$TED2_i^{(t)}$	Automation potential index	Substitutability of tasks by automation	index
$TED3_i^{(t)}$	Technology investment intensity	IT / capital investment intensity	USD/year
$TED4_i^{(t)}$	Workflow standardization level	Ease of systematization and process codification	index

表 2: Core notations used in Sub-Model I.

Symbol	Descriptions	Unit
N	The number of careers (occupations)	count
m	The number of indicators	count
n	The number of years in the panel window	year
K	The number of clusters (archetypes)	count
L	Look-back length used to form clustering features (recent years)	year
H	Forecast horizon for archetype trend projection	year
$x_{ij}^{(t)}$	Raw indicator value of career i on indicator j in year t	indicator-dependent
$z_{ij}^{(t)}$	Normalized indicator after direction harmonization and min-max scaling	—
w_j	Entropy-derived indicator weight	—
$\text{score}_i^{(t)}$	Annual composite career score, $\sum_{j=1}^m w_j z_{ij}^{(t)}$	—
$\mathbf{u}_i$	Recent-score feature vector for clustering, $\mathbf{u}_i \in \mathbb{R}^L$	—
$\mathcal{C}_k$	Set of careers assigned to archetype (cluster) k	—
$\boldsymbol{\mu}_k$	Cluster centroid in the feature space	—
$\overline{\text{score}}_k^{(t)}$	Archetype mean score (cluster mean) in year t	—

表 3: Indicator system for Sub-Model I (with units and direction). (To be inserted)

composite score of occupation i in year t is defined as

$$\text{score}_i^{(t)} = \sum_{j=1}^m w_j z_{ij}^{(t)}. \quad (2)$$

By aggregating the annual scores, we obtain the multi-year composite score trajectory for career i :

$$\text{score}_i^{(t_1 \sim t_n)} = [\text{score}_i^{(t_1)}, \text{score}_i^{(t_2)}, \dots, \text{score}_i^{(t_n)}]. \quad (3)$$

This trajectory reflects both the overall level of career development and its evolution over time. The weights obtained in this calculation are shown in Figure 4. They not only provide an intuitive view of which indicator dimensions are the primary drivers of cross-career differences in the sample, but also offer an interpretable basis for the subsequent clustering and forecasting analyses.

图 4: Entropy-derived indicator weights w_j (higher values indicate stronger discriminatory power). (To be inserted)

Careers are clustered based on the similarity of their recent score trajectories in order to identify typical occupational archetypes. Specifically, each career i is transformed into a compact feature vector composed of its most recent L annual scores:

$$\mathbf{u}_i = [\text{score}_i^{(t_n-L+1)}, \dots, \text{score}_i^{(t_n)}] \in \mathbb{R}^L, \quad (4)$$

so that careers with similar development levels and trends lie closer to one another in this L -dimensional space.

K-means clustering is then applied to the set of feature vectors $\{\mathbf{u}_i\}$, partitioning them into K groups by minimizing within-cluster dispersion and thereby producing directly usable labels for career archetypes:

$$\min_{\{\mathcal{C}_k\}_{k=1}^K} \sum_{k=1}^K \sum_{\mathbf{u}_i \in \mathcal{C}_k} \|\mathbf{u}_i - \boldsymbol{\mu}_k\|_2^2, \quad (5)$$

where $\mathcal{C}_k$ and $\boldsymbol{\mu}_k$ denote cluster k and its centroid, respectively. The number of clusters K is selected using elbow and silhouette diagnostics to strike a balance between parsimony, meaning fewer and more interpretable archetypes, and fidelity, meaning sufficient separation among distinct score-trajectory patterns.

为了直观展示高维指标空间中的聚类结构, 本文采用二维 t-SNE 嵌入对样本进行可视化, 相关结果见图5。需要说明的是, t-SNE 仅用于展示, 不会参与也不会改变 K-means 的聚类划分。图中三类点云在二维平面上呈现出清晰的分离状态, 这说明综合指标空间中确实存在稳定的职业原型, 而非单一的连续分布。为了让结果更易理解, 本文为每个簇选取了一个代表性职业作为锚点: Cluster 0 对应 Trade 主导的原型, 以铁路维修工为代表; Cluster 1 对应 STEM 主导的原型, 以管理分析师为代表; Cluster 2 对应 Arts 主导的原型, 以平面设计师为代表。结合各簇画像总结, Cluster 1 在时间序列上保持着较高的综合得分, 同时有着更强的需求信号和更好的教育回报; Cluster 0 的综合得分整体偏低, 还呈现出更高的自动化暴露程度和较慢的技能更新特点; Cluster 2 的得分水平处于中间位置, 但分布的离散度和

波动性更为显著，能看出更高的不确定性和收益分化情况。总体来看，图5 为不同职业家族在 Gen-AI 冲击下具有显著异质性这一结论提供了直观证据，也为后续按簇开展趋势预测和策略建议奠定了基础。

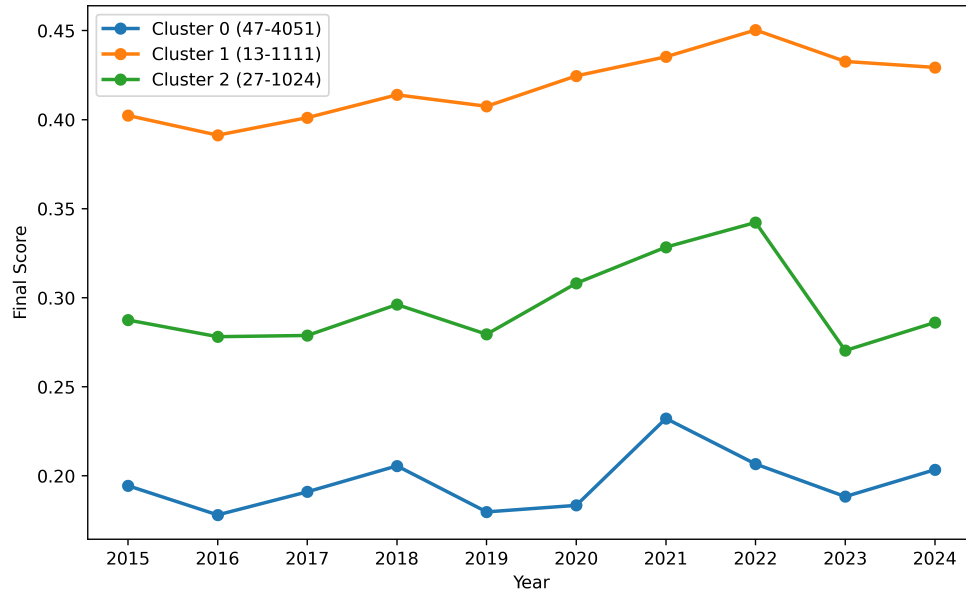


图 5: Two-dimensional t-SNE visualization of careers colored by K-means cluster labels.

5.1.3 Results: Archetype Interpretation, Representative Careers, and Trend Projection

本小节将聚类输出转化为能够直接服务于研究目标的结果。由于研究问题已经预先给出了三个广义类别标签——理工（STEM）、职业技术（Trade）和艺术（Arts）——我们首先检验聚类结果是否与这一分类体系具有合理的一致性。针对若干个可能的聚类数 K ，我们计算调整兰德指数（Adjusted Rand Index），并以轮廓系数（silhouette coefficient）衡量类间分离度，然后在可解释性与聚类经验质量之间进行权衡，从而选取最终的 K 值。我们特别强调，适中的调整兰德指数是理想的结果，因为这表明所提取的职业原型捕捉到的结构模式与原有标签相关，但并非对其的简单复制。

表 4: Model selection summary for K : ARI (vs. STEM/Trade/Arts), silhouette, and notes. (To be inserted)

K	ARI	Silhouette	Notes
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Once K is selected, each cluster is interpreted by combining its centroid-level indicator profile with its score trajectory relative to the overall distribution. In practice, the cluster mean trajectory $\overline{\text{score}}_k^{(t)}$ is used to describe the baseline level and recent direction of change, while the strongest contributing dimensions are used to diagnose whether the archetype is primarily characterized by demand-side strength, return stability, skill structure, or technology dependence. This interpretation step turns the statistical groupings into decision-relevant narratives, clarifying for each archetype whether it appears more vulnerable to task substitution or more likely to benefit from Gen-AI-enabled augmentation.

表 5: Career archetypes: interpretation summary, cluster size, and representative occupation. (To be inserted)

Cluster	Interpretation (key traits and Gen-AI exposure)	Size	Representative occupation
C1	[TBD]		[TBD]
C2	[TBD]		[TBD]
C3	[TBD]		[TBD]

Finally, Sub-Model I projects medium-term trends at the level of occupational archetypes, which is more stable and less error-prone than forecasting each occupation individually. For each cluster k , the archetype score series is defined as the cluster mean:

$$\overline{\text{score}}_k^{(t)} = \frac{1}{|\mathcal{C}_k|} \sum_{i \in \mathcal{C}_k} \text{score}_i^{(t)}.$$

(6)

In view of the potential trend shift induced by the emergence of generative AI (with technical details omitted here), a parsimonious segmented trend model is fitted to this series, and bootstrap methods are used to construct uncertainty bands for forecasts over the next H years. The resulting projections, combined with the interpretation of each occupational archetype, provide a clear and traceable basis for selecting three target careers for in-depth, institution-specific analysis, with exactly one career chosen from each of the STEM, trade, and arts categories.

图 6: Forecasted archetype score trajectories with uncertainty bands. (To be inserted)

5.2 Sub-Model II:

5.3 Sub-Model III:

6 Results and Extension

6.1 Optimization Objectives

6.2 Constraints Analysis

6.3 Optimization Result Analysis

6.4 Key Parameter Determination

7 Sensitivity Analysis

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8 Strengths and Weaknesses

8.1 Strengths

8.2 Weaknesses

9 Conclusion

Appendices

附录 A First appendix

In addition, your report must include a letter to the Chief Financial Officer (CFO) of the Goodgrant Foundation, Mr. Alpha Chiang, that describes the optimal investment strategy, your modeling approach and major results, and a brief discussion of your proposed concept of a return-on-investment (ROI). This letter should be no more than two pages in length.

Dear, Mr. Alpha Chiang

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Sincerely yours,

Your friends

Here are simulation programmes we used in our model as follow.

Input matlab source: Source file ./code/mcmthesis-matlab1.m not found.

附录 B Second appendix

some more text **Input C++ source:** Source file ./code/mcmthesis-sudoku.cpp not found.

Report on Use of AI

1. Bing AI

Query1: write a sort algorithm

Output: Sure, I can help you with that. Here is a simple implementation of the selection sort algorithm in Python:

```
def selection_sort(arr):
    for i in range(len(arr)):
        min_idx = i
        for j in range(i + 1, len(arr)):
            if arr[min_idx] > arr[j]:
                min_idx = j
        arr[i], arr[min_idx] = arr[min_idx], arr[i]
    return arr
```

This algorithm sorts an array by repeatedly finding the minimum element from the unsorted part of the array and putting it at the beginning. The time complexity of this algorithm is $O(n^2)$.

There are many other sorting algorithms available such as bubble sort, insertion sort, merge sort, quick sort, heap sort, and more. Each of these algorithms has its own advantages and disadvantages. You can learn more about them on [GeeksforGeeks](#) and [Programiz](#).

I hope this helps!

2. OpenAI Enie (Nov 5, 2023 version, Ernie 4.0)

Query1: <insert the exact wording of any subsequent input into the AI tool>

Output: <insert the complete output from the second query>

3. Github CoPilot (Feb 3, 2024 version)

Query1: <insert the exact wording you input into the AI tool>

Output: <insert the complete output from the AI tool>

4. Google Bard (Feb 2, 2024 version)

Query: <insert the exact wording of your query>

Output: <insert the complete output from the AI tool>

5. Bing AI

Query: write a FFT algorithm

Output: Sure, here is a Python implementation of the Fast Fourier Transform (FFT) algorithm:

```
import numpy as np

def fft(x):
    n = len(x)
    if n == 1:
        return x
    even = fft(x[0::2])
```

```
odd = fft(x[1::2])
T = [np.exp(-2j * np.pi * k / n) * odd[k] for k in range(n // 2)]
return [even[k] + T[k] for k in range(n // 2)] + [even[k] - T[k] for k in range(n // 2)]
```

The FFT is a widely used algorithm for computing the Discrete Fourier Transform (DFT) of a sequence. It reduces the number of computations needed from $O(n^2)$ to $O(n \log n)$ 12. The basic idea of the FFT is to apply divide and conquer. We divide the coefficient vector of the polynomial into two vectors, recursively compute the DFT for each of them, and combine the results to compute the DFT of the complete polynomial 3.

I hope this helps!